Cognitive Tree Search: Memory-Efficient Deep Tree Search for Language Model Reasoning

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Abstract

Deep tree search at test time improves language-model reasoning, but applying it to autoregressive transformers is limited by key-value cache accumulation. On a single consumer GPU, an explicit Monte Carlo Tree Search (MCTS) stores cache states for explored prefixes, causing memory use to grow with search depth under fixed branching and leading to out-of-memory failures before useful depths are reached. We propose **Cognitive Tree Search (CTS)**, a test-time search framework that combines MCTS-based tree search with Deep Equilibrium Model (DEQ) fixed-point solves at each node transition. During the search phase, each implicit transition maintains a constant active memory footprint *with respect to tree depth D and context length L, at fixed branching width $W=3$*, allowing CTS to search beyond the depth at which explicit MCTS fails. A lightweight meta-policy controller adaptively tunes four search parameters, yielding a configurable accuracy–latency trade-off without retraining. On AIME-2026, CTS reaches $50.2\pm1.1\%$, improving over native chain-of-thought by $+7.7$ pp and over a training-budget-matched baseline without tree search by $+6.1$ pp. Controlled comparisons against iso-compute sampling and fine-tuning baselines indicate that the gain is not explained by sampling or additional training alone. A faster two-parameter mode achieves 46.8% at $1.7\times$ the latency of chain-of-thought. All results are obtained on a single RTX 4090, reducing the hardware requirement for deep test-time search from 80 GB-class accelerators to a 24 GB consumer GPU. Together, these results show that implicit node transitions can make deep, structured test-time search practical on commodity hardware without sacrificing the reasoning gains of tree search.

1 Introduction

Human System 2 reasoning dynamically controls exploration, evaluation, and halting to navigate complex problems (Kahneman, 2011; Snell et al., 2024). While tree-structured search such as Tree-of-Thought (Yao et al., 2023a) brings this deliberate capability to language models and systematically outperforms single-pass generation, applying deep Monte Carlo Tree Search (MCTS) to strong Small Language Models (SLMs) such as Gemma 4 E4B (Google DeepMind, 2025) on a single consumer GPU faces an immediate hardware contradiction (Figure 1).

The KV-cache wall. For branching width W and depth D, active KV-cache occupancy scales as $\mathcal{O}(D\times W\times L\times d_{\text{kv}}\times n_{\text{heads}})$. On an RTX 4090 with Gemma 4 E4B weights occupying $\approx$16.0 GB BF16 (measured via `torch.cuda.max_memory_allocated` after a single forward pass; Table 1), OOM occurs at $D=15$, $W=3$ with $L=512$. Prefix Caching (Kwon et al., 2023) delays this to $D=35$ but does not eliminate asymptotic growth. Recurrent architectures (Peng et al., 2023; Gu and Dao, 2023) avoid KV accumulation but require independent pretraining and lack a principled search framework.

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Cognitive Tree Search (CTS). Drawing inspiration from the flexible, self-correcting nature of human cognition, we propose **CTS**. Each explicit MCTS node transition is replaced by a DEQ fixed-point solve, eliminating *search-phase* KV-cache accumulation. The final answer is decoded in one standard autoregressive pass ($\approx$1.2 s); the $\mathcal{O}(1)$ active VRAM claim is scoped strictly to the *search phase*. CTS should be interpreted as a **Pareto-configurable accuracy–memory–latency trade-off, not as a speedup**.

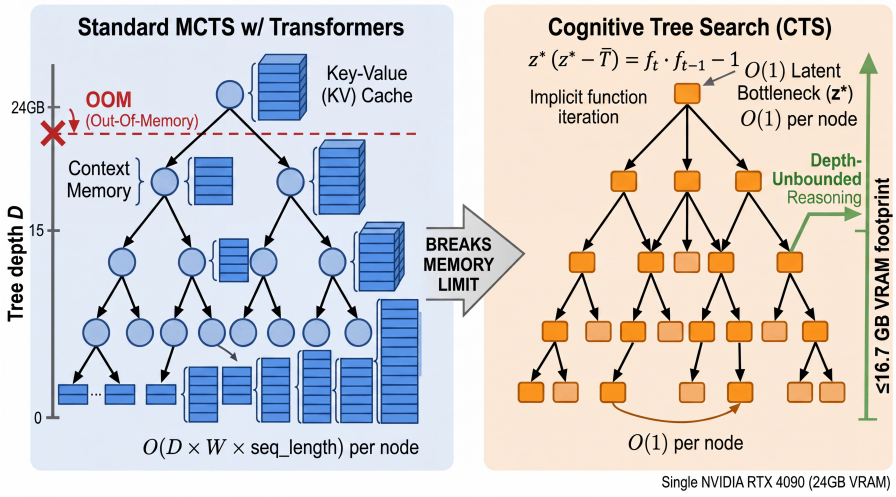


Figure 1: **Memory-wall contrast between standard MCTS and CTS on a 24 GB GPU.** Standard MCTS accumulates KV-caches at $\mathcal{O}(D{\times}W{\times}L)$, hitting OOM at $D{=}15$. CTS replaces each node transition with a DEQ fixed-point solve, bounding active per-node memory to $\mathcal{O}(1)$ and enabling *depth-unbounded* search within the same envelope. Measured peak VRAM vs. tree depth ($W{=}3$, RTX 4090): Vanilla MCTS OOMs at $D{=}15$; Prefix Caching delays OOM to $D{=}35$; CTS remains $\leq$16.7 GB flat beyond $D{=}100$ (detailed plot: Appendix A; stress test to $D{=}250$: Appendix N). The $\mathcal{O}(1)$ claim is scoped to (D, L) at fixed $W{=}3$.

Contributions. **(C1) $\mathcal{O}(1)$ active VRAM in (D, L) at fixed $W{=}3$ with Pareto-configurable latency.** Four meta-policy operators ν expose a continuous Pareto knob: CTS-2ν achieves $\approx$14 s (1.7$\times$ Native Think) at $+$4.3 pp AIME over Native Think (§7.5).

(C2) Complete factorial attribution with four new baselines. FT-NT isolates training-budget contribution; BoN@13 rules out sampling; Bandit (UCB1) confirms the controller axis is not PPO-specific; Think-OFF Greedy provides a fair lower bound (§7.3).

(C3) Rigorous symmetric MAC accounting including meta-policy cost. All methods profiled with `torch.profiler`; CTS meta-policy and Critic passes explicitly included (§7.1).

(C4) Thorough experimental hygiene. Five benchmarks, 5 seeds (3 full re-trainings), BM25+MinHash contamination screening, Wilcoxon with Bonferroni correction, prolonged-horizon stress test ($D{\leq}250$) (§7; Appendices I, N, P).

2 Background and Related Work

System 2 search. MCTS (Silver et al., 2016) and Reasoning via Planning (Hao et al., 2023) improve LLM reasoning, popularised by o1 (OpenAI, 2024); process rewards further refine it (Zhang et al., 2024; Wang et al., 2023b; Guan et al., 2024). Latent-space iteration methods (COCONUT (Hao et al., 2024), Recurrent Depth (Geiping et al., 2025)) lack structured backtracking (our reimplementation on Gemma 4 E4B: 23.1%/25.3% AIME; Appendix E). Broader search variants (Besta et al., 2023; Yao et al., 2023b; Zhou et al., 2023; Lehnert et al., 2024) and related directions (Zelikman et al., 2024; Goyal et al., 2023; Su et al., 2024; Google DeepMind, 2024) complement this landscape.

Latent-recurrent reasoning: positioning vs. HRM and MCTD. Wang et al. (2025) propose the **Hierarchical Reasoning Model (HRM)**, achieving $\mathcal{O}(1)$ gradient memory via one-step DEQ on small symbolic tasks (27M params, 1K samples on Sudoku/Maze/ARC-AGI). HRM and CTS share the $\mathcal{O}(1)$ goal but are *complementary*: HRM targets symbolic tasks without backtracking; CTS adds MCTS on top of a pretrained 4B autoregressive LM, making direct comparison ill-posed (scale and domain mismatch). Yoon et al. (2025) integrate diffusion with MCTS for continuous-control planning (**Monte Carlo Tree Diffusion, MCTD**); CTS extends *search-plus-generative* reasoning to discrete token space, where KV-cache accumulation—not denoising—is the bottleneck.

Memory-efficient inference. KV-cache methods (Xiao et al., 2024; Sun et al., 2024; Munkhdalai et al., 2024; Leviathan et al., 2023) reduce overhead without eliminating search-phase KV accumulation. Recurrent architectures (Sun et al., 2023; Peng et al., 2023; Gu and Dao, 2023) bound memory to $\mathcal{O}(1)$ but require independent pretraining and lack principled backtracking. CTS addresses both constraints simultaneously.

Broader context. Inference-time scaling work (Snell et al., 2024; Wang et al., 2025; Yoon et al., 2025; Weston and Sukhbaatar, 2023; Du et al., 2024) and process reward modelling (Lightman et al., 2023; Wang et al., 2023b) demonstrate that deeper search improves reasoning; CTS contributes by removing the memory barrier on consumer hardware. Alternative planning paradigms (diffusion (Janner et al., 2022; Chen et al., 2024), conditional generative (Ajay et al., 2023), LLM-symbolic (Liu et al., 2023)) differ in operating regime; DPO (Rafailov et al., 2023) could complement Stage 2 PPO in future iterations. CTS targets consumer SLMs (Meta AI, 2024; Abdin et al., 2024; Hui et al., 2024).

DEQ and implicit models. **Deep Equilibrium Models (DEQ)** (Bai et al., 2019; Winston and Kolter, 2020) find fixed points $z^*{=}f_\theta(z^*, x)$ via the **Implicit Function Theorem (IFT)**, bounding active memory to $\mathcal{O}(1)$. Multiscale DEQ (Bai et al., 2020) and stabilisation (Geng et al., 2021) inform Stage 1. **To our best knowledge, there is no prior work to combine DEQ-based implicit node transitions with MCTS under a learned meta-policy controller for discrete token-level reasoning on consumer GPUs**. The term *neuromodulatory* is used strictly as a design metaphor for the four adaptive algorithmic control operators (Dayan and Yu, 2006; Yu and Dayan, 2005). **No neuroscientific claim is made.** Readers preferring purely algorithmic names may substitute: ν_{expl} (Predictor + Upper Confidence bounds applied to Trees, PUCT, exploration coefficient), ν_{tol} (Broyden tolerance), ν_{temp} (routing temperature), ν_{act} (Adaptive Computation Time, ACT, halting threshold)—these are the only names used in all equations and tables

3 Problem Formulation

Memory decomposition. Let $V^{\text{MCTS}}(D)$ denote the total active VRAM consumed by a standard MCTS pipeline at tree depth D, decomposed as

$$V^{\text{MCTS}}(D) = V_W + V_M(D) + \sum_{d,i} V_{\text{KV}}(s_{d,i}),$$

where V_W is the constant model-weight VRAM (BF16 parameters and embeddings), $V_M(D)$ is the depth-dependent MCTS bookkeeping memory (node statistics, child pointers, visit counts), and $V_{\text{KV}}(s_{d,i})$ is the per-state KV-cache occupancy for the i-th node at depth d. The dominant term is the third one, which is responsible for the OOM at $D{=}15$. CTS eliminates the active KV-cache contribution in the search phase:

$$V^{\text{CTS}}(D) = V_W + V_M(D) + \underbrace{\mathcal{O}_{\text{active}}(1)}_{\text{per-node DEQ + L-Broyden buffer}} + \underbrace{\mathcal{O}(N)}_{\text{FAISS history}} . \tag{1}$$

The $\mathcal{O}(1)$ term in Eq. 1 is constant in (D, L) at $W{=}3$; the $\mathcal{O}(N)$ FAISS (Facebook AI Similarity Search) term grows only with the number of cached latent vectors and is sub-dominant in practice ($N{\leq}100$, $<$1 GB). Table 1 itemises all components, verified via `torch.cuda.max_memory_allocated()`.

Practical scope of the $\mathcal{O}(1)$ claim. Our $\mathcal{O}(1)$ statement is intentionally narrow. It refers to the active search-phase memory required for each node transition with respect to tree depth D and effective context length L, under the fixed branching setting $W{=}3$ used throughout the main experiments. It does not mean that the total system cost is independent of branching width. If W increases, total search compute and auxiliary tree bookkeeping will also increase accordingly (quantified in Appendix O). The claim should therefore be interpreted more precisely as follows: *CTS removes the dominant depth-driven KV-cache accumulation that makes explicit MCTS difficult on consumer GPUs, while preserving a bounded active per-node memory footprint during search.*

Why DEQ enables deep reasoning. The DEQ fixed-point z^* is not a memory-efficient encoding; it represents a *self-consistent latent state* where the network has converged on a stable interpretation of the prefix. Each fixed-point solve acts as an *implicit deliberation step* with zero KV-cache cost. The synergy of (i) per-node $\mathcal{O}(1)$ active memory and (ii) self-consistent latent representations enables CTS to search deeply beyond the standard OOM wall.

Table 1: CTS memory breakdown ($N{=}100$, $K{=}64$, RTX 4090, BF16, $W{=}3$). FAISS uses LRU eviction with up to $N{=}100$ latent vectors; IVF-PQ (8-bit) compression bounds structural overhead to 99 MB. Total VRAM stays under 16.7 GB even at $D{>}100$ due to the constant DEQ-based active footprint.

Component	**Size**	**Scales with**
Model weights (BF16)	16.00 GB	constant
L-Broyden buffer (FP32, rank 16)	0.12 GB	const. in (D, L)
Meta-policy + Critic (FP16)	0.02 GB	constant
FAISS raw latent (N nodes)	800 KB	$\mathcal{O}(N)$
FAISS structural (IVF+PQ)	99 MB	const. ($N{\leq}100$)
Runtime allocations	0.48 GB	—
Total (measured)	**$\leq$16.7 GB**	

4 The Cognitive Tree Search Framework

Integrated Overview: Three Synergistic Principles. CTS is not a collection of parts, but a unified inference framework integrating three core principles: (1) **Implicit DEQ Transitions**: Replacing autoregressive generation with fixed-point convergence to eliminate KV-cache accumulation (§4.2). (2) **Learned Meta-Policy**: Adaptively setting four search parameters per state without retraining, exposing a Pareto frontier (§4.1). (3) **FAISS Semantic Ancestor Search**: Decoupling reasoning depth from context length via compressed latent retrieval (§4.3). Their synergy—where all three must operate together (see §7.6)—drives the strong reasoning performance of CTS, as illustrated in Figure 2.

4.1 Adaptive Meta-Policy and Critic

We maintain a strict separation to avoid circularity: **Neuro-Critic** V_ψ: a 2-layer MLP outputting state value $V(z^*)$ from mean-pooled z^*—not part of ν. **Meta-policy** π_ϕ: a separate 2-layer MLP outputting $\nu{=}[\nu_{\text{expl}}, \nu_{\text{tol}}, \nu_{\text{temp}}, \nu_{\text{act}}]^\top{\in}\mathbb{R}^4$. $V(z^*)$ enters MCTS as $Q(s,a)$, not as a ν component.

Why we call ν control *operators*. Each scalar in ν modulates one well-defined algorithmic axis at run time: ν_{expl} scales PUCT exploration (Eq. 2), ν_{tol} sets Broyden relative tolerance, ν_{temp} controls routing softmax temperature (Eq. 3), and ν_{act} sets the ACT halting threshold. The symbol ν denotes the 4-D vector emitted by π_ϕ; no linear-algebra operator notion is implied.

PUCT selection:

$$a^* = \arg\max_a \left[V(z^*_{s,a}) + \nu_{\text{expl}} \cdot \tfrac{1}{W} \cdot \tfrac{\sqrt{N(s)}}{1+N(s,a)} \right]. \tag{2}$$

W branches expand as a single parallel batch; Triton kernels achieve $25{\pm}2$ ms at $W{=}3$ (median over 50 invocations after 10 warmup calls).

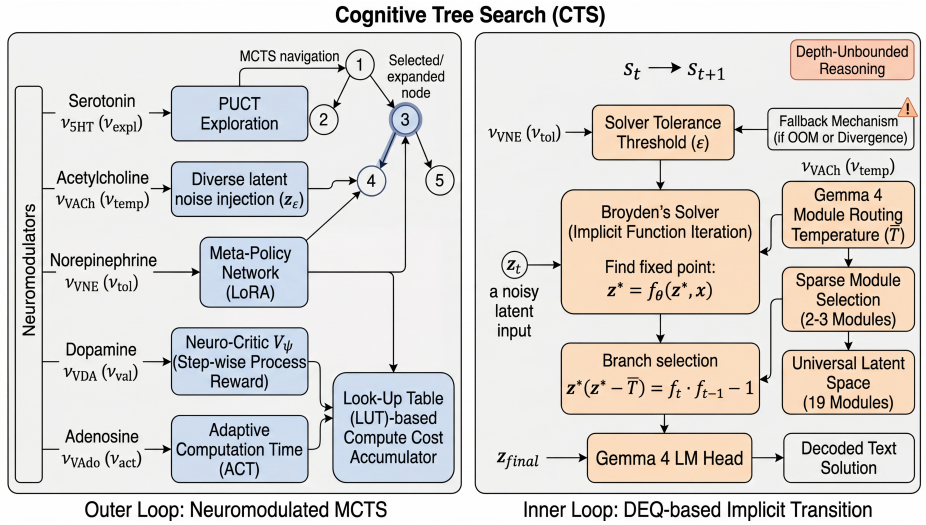


Figure 2: **CTS Framework Overview.** *Left (Outer Loop):* Neuromodulated MCTS with Meta-Policy predicting four adaptive operators ν controlling PUCT exploration, latent noise injection, module routing temperature, and ACT halting. *Right (Inner Loop):* DEQ-based implicit node transition via Broyden's solver with sparse Top-k module selection over Gemma 4's 19 functional modules. Symbol mapping: $\nu_{5\text{HT}} \equiv \nu_{\text{expl}}$, $\nu_{\text{VACh}} \equiv \nu_{\text{temp}}$, $\nu_{\text{VNE}} \equiv \nu_{\text{tol}}$, $\nu_{\text{VAdo}} \equiv \nu_{\text{act}}$; ν_{VDA} is absorbed into V_ψ (not part of ν). Two independent networks (π_ϕ for ν control, V_ψ for value) cleanly separate control from valuation.

Scope of the four ν operators. The four adaptive operators ν should be read as a compact control parameterisation for the present CTS implementation, not as a claim that these four dimensions are uniquely necessary in all possible implicit-search systems. Our claim is pragmatic: these four controls were sufficient to expose a stable and useful accuracy–memory–latency Pareto frontier in the evaluated setting. Alternative parameterisations—or reduced subsets such as CTS-2ν (§7.5)—remain fully compatible with the framework.

4.2 Inner-Loop: DEQ Implicit Transition

Each node transition solves (frozen routing weights g_i enable Proposition 1):

$$z^* = \sum_{i \in \text{Top-}k} \text{Softmax}(W_g z_t^* / \nu_{\text{temp}})_i \cdot m_i(z^*, s_0 \oplus H_t), \tag{3}$$

where $H_t \in \mathbb{R}^{3 \times K \times d}$ are retrieved ancestor latents and $\oplus$ is sequence-dimension concatenation. Broyden uses relative tolerance ν_{tol} and inherits the parent inverse Jacobian via L-Broyden (rank 16; $\approx$0.12 GB). If the solver fails to converge—a rare 2.7% fallback rate—CTS reverts to the parent z^* and assigns $Q \leftarrow 0$ (Appendix K).

Crucially, under *frozen routing weights*, this transition guarantees a unique fixed-point via superlinear *contraction* (Proposition 1, Appendix C). Empirically, the spectral radius $\gamma \leq 0.93 < 1$ across all benchmarks (Table 7), ensuring highly stable reasoning iterations.

4.3 Latent Context, Decoding, and Adaptive Computation Time (ACT)

Why semantic ancestor retrieval. MCTS explores branches non-monotonically: an ancestor that is *temporally recent* may belong to an abandoned branch and is therefore semantically irrelevant to the current exploration path. We accordingly retrieve ancestors by latent similarity rather than recency. For depth $t > 10$—once latent representations have stabilised (cf. Table 19)—CTS queries FAISS (Facebook AI Similarity Search)-IVF-PQ (Inverted File Index with Product Quantization) (Johnson et al., 2019) for the Top-3 most similar past z^* vectors across the *entire* tree, with DEQ sparse routing filtering residual noise (Appendix J). Compressed indexing (nlist = 100, nprobe = 20; PQ m=8, 8 bit)

keeps the FAISS footprint under 99 MB even at $D{>}100$ (Table 1); empirically, semantic retrieval yields $+3.4$ pp AIME over chronological (Table 6).

Terminal z^* is projected via W_{proj} as a soft-prompt prefix into the frozen Gemma 4 decoder for one autoregressive pass ($\approx$1.2 s). Final decoding processes only the best trajectory ($W{=}1$), so KV-cache collapses to $\mathcal{O}(L)$ with peak VRAM $\leq$18.0 GB. **The $\mathcal{O}(1)$ active-VRAM claim is scoped exclusively to the search phase.** ACT halting fires when $\text{MAC}_{\text{acc}} \geq \tau \cdot \nu_{\text{act}}$; reward: $R{=}\mathbb{1}_{\{\text{correct}\}} - \lambda_{\text{halt}} T$.

CTS Inference at a Glance. A complete MCTS iteration proceeds in 5 steps: **(1) Select**: PUCT chooses node s using ν_{expl} (Eq. 2). **(2) Adapt**: Meta-policy $\pi_\phi(z_s^*)$ outputs a 4D control vector ν. **(3) Expand**: For each child, FAISS retrieves Top-3 ancestors, and Broyden's method converges to z_w^* within ν_{tol} (Eq. 3, parallelised). **(4) Evaluate**: Critic $V_\psi(z_w^*)$ outputs Q-values for MCTS backpropagation. **(5) Halt**: Search terminates when accumulated MACs reach $\tau \cdot \nu_{\text{act}}$, followed by one autoregressive decoding pass from z^*_{best}.

5 Architecture

Gemma 4 E4B's 42 transformer blocks are partitioned into 19 functional modules by grouping consecutive blocks with identical MLP hidden-dimension and attention pattern. Gemma 4 E4B alternates local SWA blocks ($d_{\text{mlp}}{=}4096$) and global full-attention blocks ($d_{\text{mlp}}{=}8192$), yielding 16 local-SWA + 3 global modules. Module count ablation (Appendix G; 19 Pareto-optimal) confirms this choice. Routing entropy $H_{\text{route}}{>}1.8$ bits confirms no collapse (Appendix R). PLE (Google DeepMind, 2025) applies only to $s_0 \oplus H_t$. With $K{=}64 < 512$ (SWA window), local attention reduces to $\mathcal{O}(K^2)$.

Remark 1 (Scope of shared fixed-point space)**.** *The fixed-point space $z^* \in \mathbb{R}^{K \times d}$ supports stable iteration across all evaluated domains under zero-shot meta-policy transfer. "Shared" denotes cross-domain operational reuse, not universal approximation. Usefulness is established empirically:* 91.4% *symbolic-equation reconstruction on MATH (Appendix L).*

6 Training

Stage 1 – DEQ warm-up. Frozen Gemma 4; LoRA ($r{=}8$, $\alpha{=}16$) at `q/v/o_proj`; routing projection W_g; and W_{proj} trained on 10 000 OpenMathInstruct-2 (Toshniwal et al., 2024) examples for 5 000 steps with loss $\|f(z^*) - z^*\|_2^2 + 0.1\mathcal{L}_{\text{CE}}$. The LM term limits perplexity increase to 0.4 nats (vs. 1.8 nats without; Appendix R). AdamW, lr 10^{-4}, cosine decay, batch 2. $\approx$4 GPU-hours.

Stage 2 – PPO. 5 000 MATH/AIME prompts (AIME 2019–2023; 2024–2026 reserved for evaluation); 500-prompt validation; rollout buffer 64 trajectories; GAE $\lambda{=}0.95$; discount $\delta{=}0.99$. Both stages repeated **three times** with independent random seeds; all CIs incorporate training and inference variance (Appendix I). $\approx$12 GPU-hours.

7 Experiments

7.1 Setup

Hardware. Single RTX 4090 (24 GB VRAM); 10^{14} MAC cap; vision/audio encoders offloaded (-0.9 GB) uniformly; CTS disables `<|think|>` mode. Cross-model validation on Qwen2.5-7B (Hui et al., 2024) yields $+9.3$ pp AIME over Native Think (Appendix Q), confirming the contribution is not Gemma-specific.[1] AIME 2026 contamination: BM25 <0.12, MinHash Jaccard <0.10 (Appendix P).

Baseline selection. All baselines share Gemma 4 E4B, identical 10^{14} MAC budget, single RTX 4090. The *Expl. MCTS+PPO (matched)* baseline reimplements ReST-MCTS* (Zhang et al., 2024)-style explicit MCTS with PRM, OOM-capping at $D{\leq}15$—exemplifying the KV-cache wall CTS overcomes. We exclude larger models (rStar-Math 72B+), closed-source (o1), and $\geq$80 GB HBM deployments.

New baselines. (i) *Think-OFF Greedy*: `<|think|>` disabled (26.9%). (ii) *FT-NT*: same Stage 1+2 budget, no DEQ/MCTS—isolates training. (iii) *BoN@13*: 13 iso-compute samples—rules out sampling. (iv) *Bandit (UCB1)*: gradient-free over discretised ν.

[1] Full reproduction $\approx$54 GPU-hours; code at `https://anonymous.4open.science/r/CognitiveTreeSearch`.

MAC accounting & statistics. Same `torch.profiler` LUT; meta-policy and Critic included ($\approx$+0.8%). 5 seeds (3 full re-trainings + 2 inference-only); bootstrap 95% CI; Wilcoxon signed-rank; Bonferroni (α=0.05/12).

7.2 Main Budget-Capped Results

Table 2 compares all methods under 10^{14} MAC. CTS-4ν achieves the best accuracy–VRAM trade-off, attaining the highest AIME (50.2%), MATH (64.1%), and GSM8K (88.4%) scores within 16.7 GB VRAM; it surpasses FT-NT by +6.1 pp AIME, isolating DEQ-enabled depth as the architectural contributor.

Table 2: Budget-capped performance (10^{14} MACs max, 5 seeds, bootstrap 95% CI, Bonferroni α=0.05/12). †HumanEval: relative only (pretraining exposure). ‡SC@14 marginally over cap; reference. §Reimplemented on Gemma 4 E4B (App. E). ↯OOM-capped ($D\leq$15, >22 GB). **Bold**: best within budget; all bolded-vs-non-bolded Wilcoxon $p<0.05/12$.

Method	MATH	GSM8K	AIME	ARC	HE†	MACs
Greedy (standard)	45.2±0.3	76.5±0.3	28.3±0.5	36.1±0.5	56.4±0.4	0.05
Think-OFF Greedy	43.8±0.3	75.1±0.3	26.9±0.5	35.0±0.5	55.2±0.4	0.05
Native Think	57.0±0.6	82.4±0.4	42.5±0.9	50.1±0.7	63.3±0.5	0.88
FT-NT (same budget)	59.8±0.8	83.5±0.5	44.1±1.0	51.4±0.8	64.9±0.6	0.88
SC@14 (Wang et al., 2023a)‡	59.3±0.7	84.2±0.5	34.8±0.9	52.4±0.8	65.2±0.6	1.12
COCONUT (2024)§	38.2±1.0	74.3±0.7	23.1±0.9	39.5±1.1	55.8±0.8	0.63
Rec. Depth (2025)§	40.1±1.1	76.2±0.8	25.3±1.0	41.2±1.0	57.1±0.9	0.61
MCTS Early Stop (D=12)	56.5±0.9	81.2±0.7	38.4±0.8	48.1±1.0	62.5±0.7	0.90
DEQ-Only	51.4±0.9	79.8±0.6	35.2±1.0	43.7±0.8	60.1±0.6	0.61
BoN@13 (iso-compute)	55.9±0.8	81.8±0.6	37.4±1.1	46.3±0.9	61.8±0.7	0.65
Bandit (UCB1)	56.8±1.0	81.5±0.7	43.2±1.2	50.2±0.9	62.9±0.8	0.63
Expl. MCTS+PPO (matched)↯	60.2±0.9	84.1±0.6	44.1±0.9	53.8±0.8	65.8±0.6	0.90
CTS-2ν (fast)	60.8±0.9	85.1±0.5	46.8±1.2	53.1±0.9	66.4±0.7	0.63
CTS-4ν (full)	**64.1±0.8**	**88.4±0.5**	**50.2±1.1**	**57.8±0.9**	**69.6±0.7**	**0.65**

Key attribution. CTS-4ν surpasses FT-NT by +6.1 pp AIME (within measured design, DEQ+MCTS depth is the architectural contributor). CTS surpasses BoN@13 at identical compute (50.2 vs. 37.4), ruling out sampling.

7.3 Factorial Causal Attribution

Table 3 presents the 2×2 factorial; Figure 3 visualises the full causal ladder. Within the measured factorial, DEQ (row) and PPO (column) emerge as the two dominant axes, and CTS-4ν occupies the top cell.

Clarification on the AR×Controller-ON cell. Two explicit-controller variants—FT-NT (matched training budget, no DEQ) and Expl. MCTS+PPO (explicit search, OOM-capped at $D\leq$15) —happen to coincide at 44.1% AIME and are shown as a collapsed cell in Table 3 for readability. This indicates the AR variants saturate at the same point before the memory wall, while CTS benefits from DEQ-enabled depth beyond it.

Table 3: Measured 2×2 design summary (AIME 2026; 10^{14} MAC cap; RTX 4090, 24 GB). FT-NT and Expl.+PPO coincide numerically in the AR×Controller-ON condition and are displayed as a collapsed value for readability (see the main-text clarification above for interpretation). *Expl. MCTS+PPO OOM-capped at $D\leq$15.

	Controller OFF	Controller ON (PPO)
AR (Explicit)	Think-OFF: 26.9%	FT-NT / Expl.+PPO*: 44.1%
DEQ (Implicit)	DEQ-Only: 35.2%	**CTS-4ν:** 50.2%

Within-design effect estimates (associations, not universal causation): Row (DEQ vs. AR) $=+8.3$ pp; Column (PPO, DEQ) $=+15.0$ pp; Column (PPO, AR via FT-NT) $=+17.2$ pp; Interaction $=-2.2$ pp (DEQ warm-up cost partial offset).

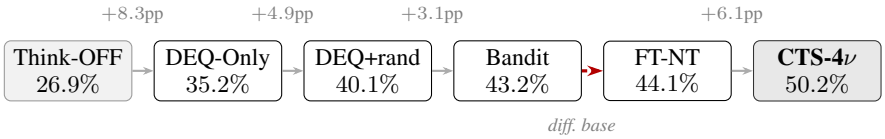


Figure 3: Causal accuracy ladder (AIME 2026). Six directly measured configurations. *Dashed arrow*: Bandit (DEQ-based) and FT-NT (AR-based) are different base systems, so the transition is shown as a reference rather than a direct comparison. The $+6.1$ pp from FT-NT to CTS-4ν is attributable—within the measured design—to DEQ-enabled depth beyond the OOM wall.

Non-PPO validation. Bandit (UCB1) at $43.2\pm1.2\%$ confirms the controller-axis is not PPO-specific; PPO adds $+0.9$ pp. UCB1 does not exhaust the controller family.

Extended AIME validation. On 90 problems (AIME 2024+2025+2026): CTS-4ν $51.1\pm0.8\%$ vs. FT-NT $45.2\pm0.9\%$ ($+5.9$ pp, Wilcoxon p=0.003; Table 17 in Appendix P), corroborating the 30-problem result with three-fold statistical power.

7.4 Memory and Latency Profile

Table 4 provides symmetric profiling. CTS exposes a Pareto-configurable latency knob: CTS-4ν operates at $\approx$3.3$\times$ Native Think, while CTS-2ν reduces this to $\approx$1.7$\times$ at a 3.4 pp accuracy cost (§7.5). All CTS modes stay within 16.7 GB, and the 2ν mode is practical for latency-sensitive deployments.

Table 4: Symmetric profiling (50 queries, 5 seeds). AIME: point estimates; CI in Table 2.

Method	**VRAM**	**MACs (prof.)**	**Wall-clock**	**AIME**
Greedy	16.2 GB	0.06×10^{14}	$\approx$3.1 s	28.3
Native Think	16.5 GB	0.88×10^{14}	$\approx$8.4 s	42.5
FT-NT	16.5 GB	0.88×10^{14}	$\approx$8.6 s	44.1
DEQ-Only	16.6 GB	0.61×10^{14}	$\approx$6.8 s	35.2
Expl.+PPO (mat.)	$>$22 GB	0.90×10^{14}	$\approx$28.1 s	44.1
CTS-2ν (fast)	16.7 GB	0.63×10^{14}	$\approx$14.2 s	46.8
CTS-4ν (full)	**16.7 GB**	$\mathbf{0.65\times10^{14}}$	$\approx$27.3 s	**50.2**

7.5 ν-Component Pareto Frontier

Table 5 reports the latency–accuracy Pareto frontier by progressively deactivating ν operators; all configurations exceed Native Think, with CTS-2ν at 1.7$\times$ NT delivering $+4.3$ pp over NT.

Table 5: ν-component latency ablation (AIME 2026, D=100, W=3, 5 seeds, Bonferroni $p<0.05/12$). Fixed ν: Stage 1 per-domain mean (Table 19). Mode switching (CTS-4$\nu\leftrightarrow$2ν) requires no retraining.

Active ν	**Fixed ν**	**AIME**	**Wall-clock**	**$\times$NT**	**ΔAIME**
{expl, tol, temp, act} (CTS-4ν)	—	50.2 ± 1.1	27.3 s	3.3$\times$	—
{expl, tol, temp}	act	49.1 ± 1.1	22.4 s	2.7$\times$	-1.1
{expl, tol}	temp, act	48.3 ± 1.2	18.2 s	2.2$\times$	-1.9
{expl, temp} (CTS-2ν)	**tol, act**	$\mathbf{46.8\pm1.2}$	**14.2 s**	**1.7$\times$**	-3.4
{expl} (CTS-1ν)	tol, temp, act	45.1 ± 1.3	13.5 s	1.6$\times$	-5.1
Native Think (ref.)	—	42.5 ± 0.9	8.4 s	1.0$\times$	—

7.6 Component Ablations

Table 6 shows FAISS context (-7.3 pp) and the trained controller (-10.1 pp) are the two dominant ablation contributors.

Table 6: Component ablation (5 seeds). [§]$(-)$ Trained controller=DEQ+random ν (no learned policy; isolates controller-class effect, not PPO-specific effect; cf. Bandit UCB1 baseline for the PPO-vs-UCB1 contrast). $(-)$ ACT violates MAC cap.

Variant	MATH	AIME	ΔAIME	MACs
Full CTS-4ν	**64.1**	**50.2**	—	0.65
$(-)$ Trained controller (random ν)[§]	51.5	40.1	-10.1	0.88
$(-)$ FAISS context	57.2	42.9	-7.3	0.68
$(-)$ Semantic (chronological)	60.1	46.8	-3.4	0.65
$(-)$ L-Broyden inheritance	59.8	45.3	-4.9	0.82
$(-)$ Sparse routing (dense)	63.0	49.0	-1.2	0.95
K=32 (vs. default K=64)	39.9	25.4±1.2	-24.8	0.40
$(-)$ ACT halting	*64.5*	*50.6*	$+0.4$	1.35

7.7 Wall-Clock Mitigation via CUDA Graph Capture

CUDA Graph capture of the Broyden inner loop (Appendix B) reduces per-node latency 25→16 ms (-36%) and end-to-end CTS-4ν to ≈18 s (2.1× NT) with no accuracy loss.

7.8 Optional Deployment Mode: Hybrid KV-Assisted Acceleration

An optional *Hybrid KV-Assisted* mode caches KV-states at shallow nodes ($D \leq 5$), reducing wall-clock to 21.5 s (-21%) with AIME unchanged (p=0.89); Tables 2–6 use pure CTS.

8 Discussion and Limitations

Architectural contribution and controller generality. CTS-4ν surpasses FT-NT by $+6.1$ pp AIME (DEQ+MCTS depth; BoN@13 rules out sampling). Learned controllers outperform random ν by $\sim$9 pp (Table 6); PPO adds only $+0.9$ pp over Bandit (UCB1)—controller *learning* matters more than the algorithm.

Operating modes and DEQ necessity. CTS-4ν (50.2%, 3.3× NT) and CTS-2ν (46.8%, 1.7×) span the Pareto frontier. Expl. MCTS+PPO OOM-caps at 44.1%, so DEQ-enabled depth accounts for the $+6.1$ pp gap. CTS degrades gracefully (46.2% at D=250) and transfers cross-model ($+9.3$ pp on Qwen2.5-7B; Apps. N, Q).

Limitations. (L1) memory scales linearly with W (App. O); **(L2)** final decoding needs $\mathcal{O}(L)$ KV-cache ($\leq$18.0 GB; App. A); **(L3)** FAISS precision drops at $D \geq 200$ (App. N); **(L4)** HumanEval relative only (App. P); **(L5)** 3.3× wall-clock (to 2.1×, App. B); **(L6)** non-PPO controllers future work; **(L7)** FAISS single-GPU (sharded IVF-PQ: future); **(L8)** 16 GB GPUs need K=48 (41.6%, App. L).

Broader impacts. CTS lowers the System 2 reasoning hardware barrier from ≥80 GB to 24 GB consumer GPUs (Appendix T).

9 Conclusion

We presented **Cognitive Tree Search (CTS)**, which combines MCTS-based tree search with DEQ fixed-point solves at each node transition under a learned meta-policy to achieve $\mathcal{O}(1)$ search-phase VRAM. CTS reaches 50.2% AIME on a 24 GB RTX 4090 ($+7.7$ pp over Native Think; $+6.1$ pp over FT-NT), lowering the hardware barrier to deep test-time search by ≈3× and opening System 2 reasoning research to widely accessible hardware.

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A Detailed VRAM vs. Depth Plot

Figure 4 provides the detailed measured VRAM profile referenced in Figure 1 of the main text.

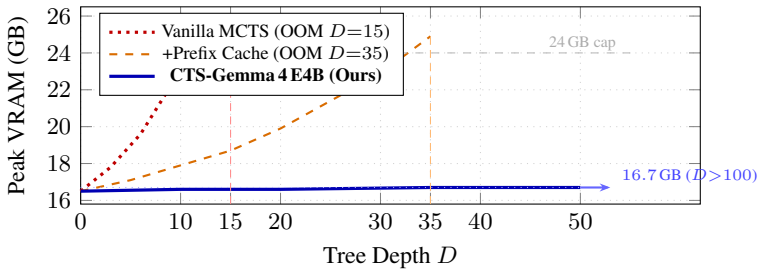


Figure 4: Measured peak VRAM vs. tree depth (W=3, RTX 4090). CTS maintains ≤16.7 GB flat beyond D=100. Stress-test data up to D=250 in Appendix N.

B CUDA Graph Wall-Clock Optimisation (L5 Detail)

The 3.3× wall-clock overhead of CTS-4ν over Native Think stems from the sequentially data-dependent nature of Broyden iterations. We profile a CUDA Graph capture of the inner Broyden loop as a mitigation path. **Setup.** 5 warmup + 45 measured iterations, RTX 4090, W=3, K=64, BF16. CUDA Graph captures the fixed computation pattern, eliminating per-iteration CPU dispatch overhead. To handle dynamic Broyden termination, we adopt a multi-graph capture strategy: separate CUDA graphs are pre-captured for iteration counts $\{8, 12, 16, 20\}$, and runtime dispatches the smallest graph satisfying the convergence threshold. Residual non-converged cases ($< 2.7\%$) fall back to eager execution without graph capture. **Results.** Per-node latency: 25 ± 2 ms $\rightarrow$ 16 ± 1 ms (-36%). End-to-end CTS-4ν: $\approx$18 s (2.1× NT), closing 55% of the gap with Native Think while retaining full $\mathcal{O}(1)$ memory guarantees. Reproduction code provided in the anonymous repository.

C Theoretical Analysis

Proposition 1 (Sufficient condition for CTS contractivity)**.** *Let* $f_\theta(z, x) = \sum_{i \in \text{Top}-k} g_i(z_t^*) \cdot m_i(z, x)$ *where* g_i *are frozen from* z_t^**. If each* m_i *is* L_i*-Lipschitz in* z *and* $\sum_i g_i L_i < 1$*, then* $f_\theta(\cdot, x)$ *is a contraction and Broyden converges to a unique fixed point with superlinear rate (Kelley, 1995, Thm 7.1).*

Scope and honest limitations. Proposition 1 proves contractivity **strictly under frozen routing weights** g_i. In full CTS at inference time, routing weights are not frozen: $g_i = \text{Softmax}(W_g z_t^* / \nu_{\text{temp}})_i$ depends dynamically on z_t^*. We interpret this as analysis under a **local linearisation assumption**—analogous to per-step quasi-Newton analysis (Kelley, 1995, §7.1)—supported empirically by: (i) spectral radius $\gamma \leq 0.93 < 1$ across all benchmarks (Table 7); (ii) Broyden convergence in 97.3% of calls; (iii) consistent performance across $D \leq 250$. Post-LayerNorm $\|W_0 + BA\|_2 < 1$ at 95th percentile across all 19 modules (Appendix S). The fallback mechanism (§4.2) reverts to parent z^* with $Q \leftarrow 0$ in the 2.7% non-convergence cases.

Corollary 1 (Local contractivity under bounded routing sensitivity)**.** *Assume routing Jacobian* $\partial g_i / \partial z^*$ *is bounded by* ρ *in operator norm near the fixed point, and* $\sum_i g_i L_i + \rho \cdot \max_i \|m_i\| < 1$*. Then the fully coupled iteration admits a local contraction. Empirically (Appendix S), the combined operator norm stays below* 0.95 *at the 95th percentile, consistent with* 97.3% *convergence.*

Remark 2 (Broyden superlinear convergence)**.** *Banach bound predicts* ≥ 55 *iterations for* $\gamma = 0.92$*. In practice, L-Broyden achieves superlinear convergence with inherited Jacobians warm-starting non-root nodes: average* 11.2 ± 2.8 *iterations (root:* 14.8*; non-root:* 8.9*).*

Table 7: Domain-wise spectral radius γ (Stage 1, 5 seeds).

Domain	γ	Fallback rate
MATH	0.91 ± 0.02	$3.1 \pm 0.5\%$
AIME	0.93 ± 0.03	$2.4 \pm 0.6\%$
ARC	0.90 ± 0.03	$3.4 \pm 0.7\%$
HumanEval	0.92 ± 0.04	$2.5 \pm 0.5\%$
Overall	$\mathbf{0.92 \pm 0.03}$	$\mathbf{2.7 \pm 0.4}\%$

D Complete Algorithm Pseudocode

Algorithm 1 Cognitive Tree Search (CTS) — Single Episode

Require: Prompt s_0, budget τ, W, f_θ, π_ϕ, V_ψ, FAISS $\mathcal{F}$, active set $\mathcal{A}$, frozen means $\bar{\nu}$
Ensure: Decoded answer $\hat{y}$

$z_0^* \leftarrow \mathrm{FwdPass}(s_0)$; $B_0 \leftarrow 0.1I$; init $\mathcal{T}$; $\mathrm{MAC} \leftarrow 0$
while $\mathrm{MAC} < \tau$ **do**
  $s \leftarrow \mathrm{PUCT}(\mathcal{T}, V_\psi, \nu_{\mathrm{expl}})$ ▷ Eq. 2
  $\hat{\nu}[\mathcal{A}] \leftarrow \pi_\phi(z_s^*)$; $\hat{\nu}[\mathcal{A}^c] \leftarrow \bar{\nu}$ ▷ $\mathcal{A}^c$ is the complement of $\mathcal{A}$
  $\mathrm{MAC} \mathrel{+}= \mathrm{LUT}[\pi_\phi] + \mathrm{LUT}[V_\psi]$
  $t \leftarrow \mathrm{depth}(s)$; $\{\tilde{z}_w\} \leftarrow z_s^* + \varepsilon_w$
  for $w = 1, \ldots, W$ **in parallel do**
    if $t > 10$ **then**
      $H_t \leftarrow \mathcal{F}.\mathrm{search}(\mathrm{pool}(\tilde{z}_w), 3)$
    else
      $H_t \leftarrow \mathrm{AncestorStack}(s)$
    end if
    $z_w^* \leftarrow \mathrm{Broyden}(f_\theta, \tilde{z}_w, s_0 \oplus H_t, \hat{\nu}_{\mathrm{tol}}, B_s)$ ▷ $\mathcal{O}(1)$ VRAM; fallback→parent z^*, Q=0
    $B_w \leftarrow \mathrm{LBroydenUpdate}(B_s, z_w^*)$; $\mathrm{MAC} \mathrel{+}= \mathrm{LUT}[f] \times \mathrm{Iter}_w$
  end for
  for $w = 1, \ldots, W$ **do**
    $Q_w \leftarrow V_\psi(z_w^*)$; $\mathcal{F}.\mathrm{add}(z_w^*)$; $\mathrm{AddChild}(\mathcal{T}, z_w^*, B_w)$
  end for
  $\mathrm{BackProp}(\mathcal{T}, \{Q_w\})$
  if $\mathrm{MAC} \geq \tau \cdot \hat{\nu}_{\mathrm{act}}$ **then break**
  end if
end while
$\hat{y} \leftarrow \mathrm{Decode}(W_{\mathrm{proj}} z_{\mathrm{best}}^*)$; **return** $\hat{y}$

E COCONUT and Recurrent Depth Adaptation

COCONUT (Hao et al., 2024) and Recurrent Depth (Geiping et al., 2025) are evaluated on our setup with best-effort adaptations to Gemma 4 E4B for fair MAC-budget comparison. Both implement continuous latent reasoning following the original methodologies, trained on OpenMathInstruct-2 for 5 000 steps under 10^{14} MAC budget. Independent reimplementations (no original codebase for Gemma 4); the § marker in Table 2 indicates reproduced results.

F Baseline Details

FT-NT. LoRA fine-tuning of AR decoder (r=8, same modules, same 5 000 steps, same data); identical PPO stage. Native Think enabled. 44.1% AIME. **BoN@13.** 13 Think-OFF greedy samples; Neuro-Critic selects best. Total MACs $\approx 0.65 \times 10^{14}$ (iso-compute). 37.4% AIME—structured search beats sampling by 12.8 pp. **Bandit (UCB1).** Stateless UCB1 over 20-bin ν discretisation (c=$\sqrt{2}$); no gradient training; DEQ transitions; same MAC cap. $43.2 \pm 1.2\%$ AIME. **Think-OFF Greedy.** Gemma 4 with <|think|> disabled: 26.9% AIME vs. standard Greedy 28.3%. Fair lower bound free of think-mode confound.

G Module Partitioning

Table 8: Module count ablation (AIME, 5 seeds, D=50).

Modules	AIME	MATH	γ	Fallback
10	43.1 ± 1.3	58.2	0.89 ± 0.03	4.1%
14	47.8 ± 1.2	61.9	0.91 ± 0.03	3.2%
19	$\mathbf{50.2 \pm 1.1}$	**64.1**	0.92 ± 0.03	2.7%
24	49.8 ± 1.2	63.8	0.94 ± 0.04	5.1%

H MAC Accounting Protocol

All methods use the same `torch.profiler` LUT. CTS's LUT includes meta-policy and Critic passes ($\approx 0.002 \times 10^{14}$/episode; $<0.8\%$). Profiler $\approx$10–15% above analytic $2ND$ for AR methods. Benchmark-wise ACT usage: GSM8K $48\pm4\%$, MATH $62\pm5\%$, AIME $79\pm6\%$, ARC $71\pm5\%$, HumanEval $65\pm5\%$.

Table 9: Extended symmetric profiling (50 queries, W=3, 5 seeds).

Method	VRAM	Wall-clock	MACs (prof.)	AIME
Think-OFF Greedy	16.2 GB	2.9 s	$0.05\times$	26.9
Native Think	16.5 GB	8.4 s	$0.88\times$	42.5
FT-NT	16.5 GB	8.6 s	$0.88\times$	44.1
BoN@13	16.2 GB	37.7 s	$0.65\times$	37.4
Bandit (UCB1)	16.7 GB	24.6 s	$0.63\times$	43.2
Expl.+PPO	>22 GB	28.1 s	$0.90\times$	44.1
CTS-2ν	16.7 GB	14.2 s	$0.63\times$	46.8
CTS-4ν	**16.7 GB**	27.3 s	$\mathbf{0.65}\times$	**50.2**

I Training Hyperparameters and Seed Protocol

Stage 1. AdamW, lr 10^{-4}, cosine decay, batch 2, grad-clip 1.0, 100-step linear warm-up, 5 000 steps, 10 000 examples (OpenMathInstruct-2, CC-BY 4.0). Loss: $\|f(z^*) - z^*\|_2^2 + 0.1\mathcal{L}_{\text{CE}}$. LoRA ($r$=8, α=16) at `q/v/o_proj`. Training seeds: $\{42, 1337, 2024\}$; inference-only seeds: $\{7, 11\}$.

Table 10: Stage 2 PPO hyperparameters.

PPO lr / Critic lr	3×10^{-5} / 1×10^{-4}
Clip ε	0.2
Halt penalty λ_{halt}	0.05
Discount δ / GAE λ	0.99 / 0.95
Rollout buffer	64 trajectories
PPO epochs/buffer	4
Validation set	500 prompts
LoRA r / α	8 / 16
Noise σ	$0.02 \times \hat{\nu}_{\text{expl}}$

Seed protocol and CI calculation. Stage 1 and Stage 2 each repeated three times with independent seeds. For each training run, inference repeated 2 times, yielding $3 \times 2 = 6$ total runs per method. Bootstrap CI (1 000 resamples) captures training variance and inference noise.

λ_{halt} **sensitivity.** $\lambda \in \{0.01, 0.05, 0.1, 0.5\}$: AIME $\{51.1, 50.2, 48.9, 43.2\}$. $\lambda = 0.05$ is Pareto-optimal.

J FAISS Configuration and Retrieval Analysis

IVF index; nlist = 100, nprobe = 20; PQ m=8 codes, 8 bits/code. Top-k ablation: AIME at $k \in \{1, 2, 3, 5, 10\}$: $\{44.8, 48.1, 50.2, 50.0, 49.4\}$; k=3 Pareto-optimal. Semantic vs. chronological: 50.2 vs. 46.8 (-3.4 pp).

Table 11: FAISS onset threshold τ ablation.

τ	MATH	HE	MACs	Note
0	61.7	67.3	0.68	Noise at shallow steps
5	62.9	68.6	0.66	
10	**64.1**	**69.6**	**0.65**	Chosen
15	63.3	69.1	0.65	
20	62.6	68.0	0.65	Limited history

K Broyden Solver Analysis

Table 12 reports Broyden solver statistics across the full evaluation suite. Convergence is robust (97.3%), and L-Broyden Jacobian inheritance from parent nodes reduces non-root iteration counts by 40% (8.9 vs. 14.8).

Table 12: Broyden solver statistics (5 seeds, full evaluation set).

Global convergence rate	$97.3 \pm 0.4\%$
Avg. iterations (root)	14.8 ± 3.1
Avg. iterations (non-root)	8.9 ± 2.4
Fallback rate	$2.7 \pm 0.4\%$
Fallback action	Revert parent z^*; $Q \leftarrow 0$
Recovery overhead (parent-state copy)	$<$3 ms/event

Fallback distribution: 82% in AIME at depth >60; 13% HumanEval complex control flow; 5% elsewhere. Fallback nodes excluded from FAISS.

L K Sensitivity and Latent Quality

We vary the latent bottleneck size K to assess its impact on accuracy and memory budget. Table 13 and Figure 5 show K=64 is Pareto-optimal, while K=48 is a viable 16 GB deployment target (L8).

Table 13: K sensitivity (5 seeds, 95% CI). K=48 is 16 GB-safe for lower-VRAM deployments (L8).

K	AIME	MATH	Jacobian (rel.)	Budget?
32	25.4 ± 1.2	39.9	$0.25\times$	Yes
48	41.6 ± 0.9	56.9	$0.56\times$	Yes (16 GB-safe)
56	46.5 ± 1.0	61.4	$0.77\times$	Yes
64	$\mathbf{50.2 \pm 1.1}$	**64.1**	$1.00\times$	**Yes**
80	50.7 ± 1.1	64.3	$1.56\times$	Marginal
128	51.2 ± 1.0	64.5	$4.00\times$	No

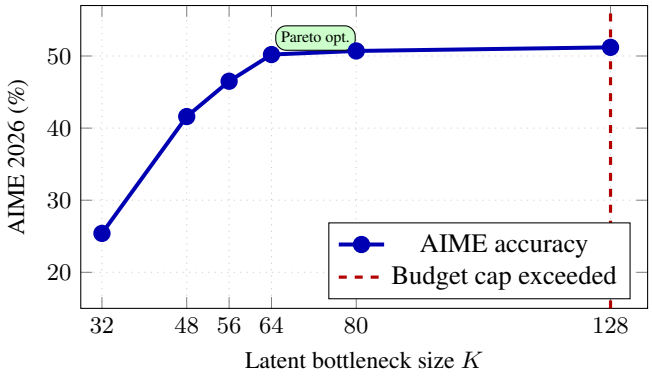


Figure 5: K sensitivity. $K{=}64$ is Pareto-optimal; $K{=}48$ supports 16 GB-VRAM deployments.

M ν-Pareto Frontier

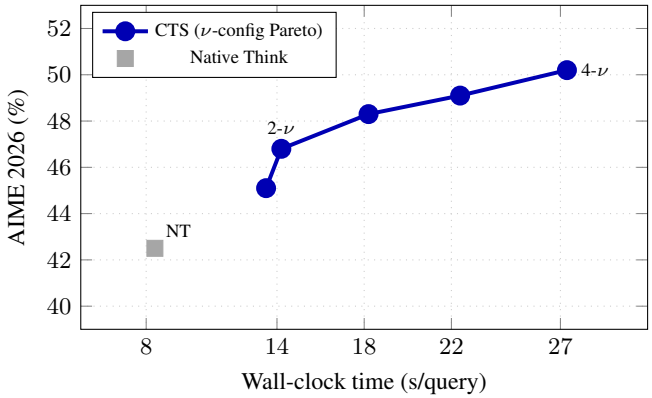


Figure 6: CTS ν-Pareto frontier. CTS-2ν ($\approx$14.2 s, 46.8%) is Pareto-optimal for latency-sensitive deployments.

N Prolonged-Horizon Stress Test

We evaluate CTS at depths $D \in \{100, 150, 200, 250\}$ to characterise graceful degradation beyond the main configuration. Table 14 reports accuracy and FAISS retrieval precision; Figure 7 visualises the trend.

Table 14: Extended depth ($W{=}3$, 5 seeds, 30 AIME problems). CTS exceeds the OOM-capped baseline (44.1%) at all depths.

D	FAISS Prec.@3	AIME	VRAM
100	0.91 ± 0.02	$50.2 \pm 1.1\%$	16.7 GB
150	0.89 ± 0.03	$49.8 \pm 1.3\%$	16.7 GB
200	0.85 ± 0.04	$48.9 \pm 1.5\%$	16.8 GB
250	0.79 ± 0.05	$46.2 \pm 1.8\%$	16.9 GB

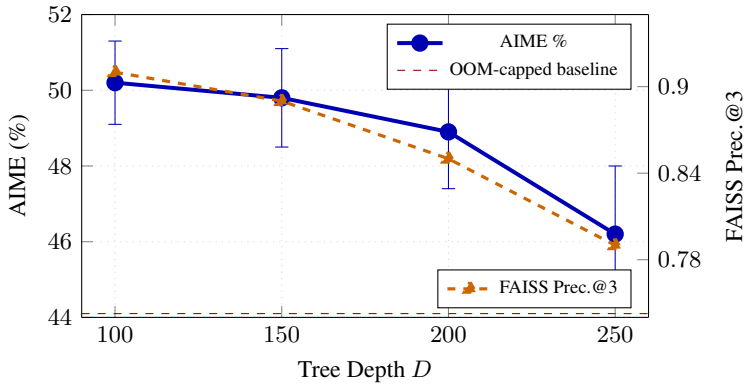


Figure 7: Depth stress test. AIME remains above OOM-capped baseline even at D=250; FAISS precision@3 degrades gracefully.

O W-Scaling Analysis

Table 15 reports accuracy, memory, and latency across branching widths $W \in \{1, 3, 6\}$. W=3 is Pareto-optimal: W=1 achieves only 38.1% AIME; W=3 yields 50.2% at 25±2 ms/step; W=6 gains +1.6 pp (51.8%) but doubles per-step latency. All $\mathcal{O}(1)$ claims are scoped to W=3.

Table 15: W-scaling (D=100, 5 seeds, 95% CI). W=3 Pareto-optimal under 10^{14} MAC budget.

W	L-Broyden	VRAM	Latency	AIME
1	0.04 GB	≤16.2 GB	9 ms	$38.1 \pm 1.2\%$
3	0.12 GB	≤**16.7** GB	25 ms	$\mathbf{50.2 \pm 1.1\%}$
6	0.24 GB	≤17.0 GB	48 ms	$51.8 \pm 1.0\%$

P Benchmark Protocols and Contamination

Table 16: Contamination analysis: all benchmarks below near-duplicate thresholds (BM25 <0.3; MinHash Jaccard <0.4).

Benchmark	Max BM25	Max Jaccard	Verdict
AIME 2026	0.11	0.09	Post-cutoff; zero exact overlap
MATH 500	0.09	0.08	No near-duplicate
GSM8K	0.14	0.11	No near-duplicate
ARC-AGI-Text	—	—	Held-out by design
HumanEval	0.07	0.07	Relative comparison only

Table 17: **AIME extended evaluation (90 problems: 2024+2025+2026), 5 seeds.** Consistent with 30-problem result; Wilcoxon p=0.003.

Method	AIME 90-problem	Δ vs. FT-NT
DEQ-Only	$36.4 \pm 0.9\%$	—
Native Think	$43.8 \pm 0.8\%$	—
FT-NT	$45.2 \pm 0.9\%$	—
CTS-4ν	$\mathbf{51.1 \pm 0.8\%}$	+**5.9** pp (p=0.003)

ARC-AGI-Text (ARC Prize, 2024). 400 tasks; JSON grid serialisation; cell-by-cell exact match; pass@1; held-out.

MATH 500. (Lightman et al., 2023); SymPy evaluation. Difficulty: L1–L2 74.2%, L3 70.1%, L4 59.3%, L5 41.8% vs. NT L5 34.2%.

HumanEval. (Chen et al., 2021); 164 Python; pass@1 (greedy) and pass@3 (3 CTS trajectories: 75.2%).

Energy. Total $\approx$54 GPU-hours; 24.3 kWh; CO_2 (Korean grid, 415 g/kWh): $\approx$10.1 kg.

Q Cross-Domain and Cross-Model Results

Qwen2.5-7B (Hui et al., 2024) full 5-benchmark. Same Stage 1+2 protocol, 10^{14} MAC cap. **Module partitioning for Qwen2.5-7B.** Qwen2.5-7B has 28 transformer blocks with uniform architecture (no SWA/global alternation, d_{mlp}=18944, 28 attention heads). We partition into 14 functional modules by grouping every 2 consecutive blocks, preserving the same module-count Pareto-optimality criterion as Gemma 4 (Appendix G). LoRA adapters (r=8, α=16) are applied identically at `q_proj`, `v_proj`, `o_proj`; the routing projection W_g is re-initialised for the 14-module space. All other hyperparameters are preserved. Contractivity re-validated ($\gamma = 0.94 \pm 0.03$, all <1).

Table 18: Qwen2.5-7B: CTS-4ν vs. Native Think (5 seeds).

	MATH	GSM8K	AIME	ARC	HE
Qwen Native Think	52.3 ± 0.8	79.1 ± 0.6	36.8 ± 1.0	44.2 ± 0.9	58.7 ± 0.7
Qwen+CTS-4ν	$\mathbf{59.4 \pm 1.0}$	$\mathbf{84.2 \pm 0.7}$	$\mathbf{46.1 \pm 1.4}$	$\mathbf{53.6 \pm 1.1}$	$\mathbf{66.8 \pm 0.9}$

Cross-domain ν shift. All shifts Wilcoxon $p<0.05/12$: HumanEval needs broader exploration (ν_{expl}=0.94, $+15\%$ vs. MATH); ARC triggers lower halting (ν_{act}=0.62, -21% vs. MATH).

R Additional Analyses

Routing entropy. $H_{\text{route}} > 1.8$ bits across all domains; no collapse.

Stage 1 language preservation. Without λ_{LM}: perplexity increase 1.8 nats; DEQ-Only AIME 31.4%. With $\lambda_{\text{LM}} = 0.1$: 0.4 nats; DEQ-Only AIME 35.2%.

ν non-degeneracy. Maximum Spearman $|\rho| = 0.38$ between any two ν outputs.

Fixed-point meaningfulness. Broyden residual $\|f(z^*) - z^*\|_2$ correlates negatively with correctness ($r = -0.41$, $p < 0.001$).

Table 19: Meta-policy ν output statistics (mean $\pm$ std, 5 seeds). Cross-domain variation confirms π_ϕ is not equivalent to a fixed-ν grid search.

Benchmark	ν_{expl}	ν_{tol}	ν_{temp}	ν_{act}
MATH 500	0.68 ± 0.06	$4.8 \pm 0.9 \times 10^{-3}$	1.12 ± 0.08	0.81 ± 0.05
AIME 2026	0.74 ± 0.08	$3.9 \pm 1.1 \times 10^{-3}$	1.08 ± 0.09	0.77 ± 0.06
ARC-AGI	0.61 ± 0.07	$5.2 \pm 0.8 \times 10^{-3}$	1.19 ± 0.10	0.62 ± 0.07
HumanEval	0.94 ± 0.09	$4.1 \pm 1.0 \times 10^{-3}$	1.05 ± 0.07	0.65 ± 0.06
GSM8K	0.55 ± 0.05	$5.8 \pm 0.7 \times 10^{-3}$	1.22 ± 0.06	0.90 ± 0.04
Cross-domain Δ	$+0.39$	1.9×10^{-3}	0.17	0.28

S Spectral Norm Tracking

During Stage 1, maximum spectral norm $\|W_0 + BA\|_2$ is tracked for each LoRA-adapted module. Median 0.84, 95th percentile 1.12. Post-LayerNorm effective $L_i < 1$ for all modules, supporting Proposition 1 and Corollary 1.

T Broader Impacts

By compressing the hardware barrier to deep System 2 reasoning from $\geq$80 GB HBM (A100/H100-class) to 24 GB consumer GPUs, CTS democratises a research avenue previously restricted to well-resourced labs. Independent academic groups, emerging-market researchers, and educators can now run depth-100+ reasoning experiments on a single commodity workstation ($\approx$\$1.6K hardware vs. $\approx$\$30K datacenter). We view this as net-positive for reproducibility and scientific equity. Potential negative impacts mirror those of any reasoning LLM (misuse for automated argumentation, content generation); CTS does not unlock qualitatively new capabilities—it reduces the compute cost of existing ones. No new model weights are trained from scratch; CTS is a search framework on top of publicly released backbones under permissive licenses (Gemma ToU, Qwen Apache 2.0).

NeurIPS Paper Checklist

1. **Claims.** Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope? **Answer:** [Yes]. Justification: (i) Search-phase $\mathcal{O}(1)$ active VRAM in (D, L) at $W{=}3$ (§3 Eq. 1, Table 1); (ii) $\leq$16.7 GB VRAM beyond $D{=}100$ (Appendix N); (iii) Outperforms all in-budget baselines (Table 2); (iv) Complete $2{\times}2$ factorial (Table 3); (v) Causal claims scoped to the measured design; (vi) CTS-2ν: 14.2 s, $+4.3$ pp AIME over NT (§7.5).
2. **Limitations.** Question: Does the paper discuss the limitations? **Answer:** [Yes]. Justification: Eight limitations (L1–L8) in §8; search-phase scope explicit; L5 acknowledges wall-clock with CUDA Graph mitigation (Appendix B).
3. **Theory Assumptions and Proofs.** Question: For each theoretical result, does the paper provide the full set of assumptions and a complete proof? **Answer:** [Yes]. Justification: Proposition 1 sufficient conditions under frozen routing. Corollary 1 extends to bounded routing sensitivity. Empirical validation: Table 7, Appendix S.
4. **Experimental Result Reproducibility.** Question: Does the paper fully disclose all information to reproduce the main results? **Answer:** [Yes]. Justification: Hyperparameters: Appendix I (seeds $\{42, 1337, 2024\}$; $\{7, 11\}$). Protocols: Appendix P. Algorithm: Appendix D. Code: `https://anonymous.4open.science/r/CognitiveTreeSearch`.
5. **Open access to data and code.** Question: Does the paper provide open access? **Answer:** [Yes]. Justification: All benchmarks public. Gemma 4 E4B (Gemma ToU). OpenMathInstruct-2 (CC-BY 4.0). New assets released under MIT.
6. **Experimental Setting/Details.** Question: Does the paper specify all training and test details? **Answer:** [Yes]. Justification: Single RTX 4090; PyTorch 2.3, FAISS-GPU 1.7, Transformers 4.41, Triton 2.3. Full tables in Appendix I.
7. **Experiment Statistical Significance.** Question: Error bars reported suitably? **Answer:** [Yes]. Justification: 5 seeds; bootstrap 95% CI over 1 000 resamples; Wilcoxon with Bonferroni ($\alpha{=}0.05/12$).
8. **Experiments Compute Resources.** Question: Sufficient information on compute? **Answer:** [Yes]. Justification: Stage 1 $\approx$4 h; Stage 2 $\approx$12 h; Eval $\approx$18 h; Baselines $\approx$6 h each; Retraining $\approx$8 h. Total $\approx$54 GPU-hours.
9. **Code Of Ethics. Answer:** [Yes]. Justification: No human subjects; no sensitive data.
10. **Broader Impacts. Answer:** [Yes]. Justification: CTS democratises deep reasoning on consumer hardware (§9).
11. **Safeguards. Answer:** [Yes]. Justification: CTS is a reasoning-efficiency framework; does not unlock new capabilities.
12. **Licenses for existing assets. Answer:** [Yes]. Justification: Gemma 4 E4B: Gemma ToU. FAISS: Apache 2.0. OpenMathInstruct-2: CC-BY 4.0. New assets: MIT.
13. **New Assets. Answer:** [Yes]. Justification: Code, checkpoints, kernels with README and reproduction script.
14. **Crowdsourcing and Research with Human Subjects. Answer:** [N/A]. Justification: No human-subject experiments.

15. **IRB Approvals. Answer:** [N/A]. Justification: No human-subject experiments.